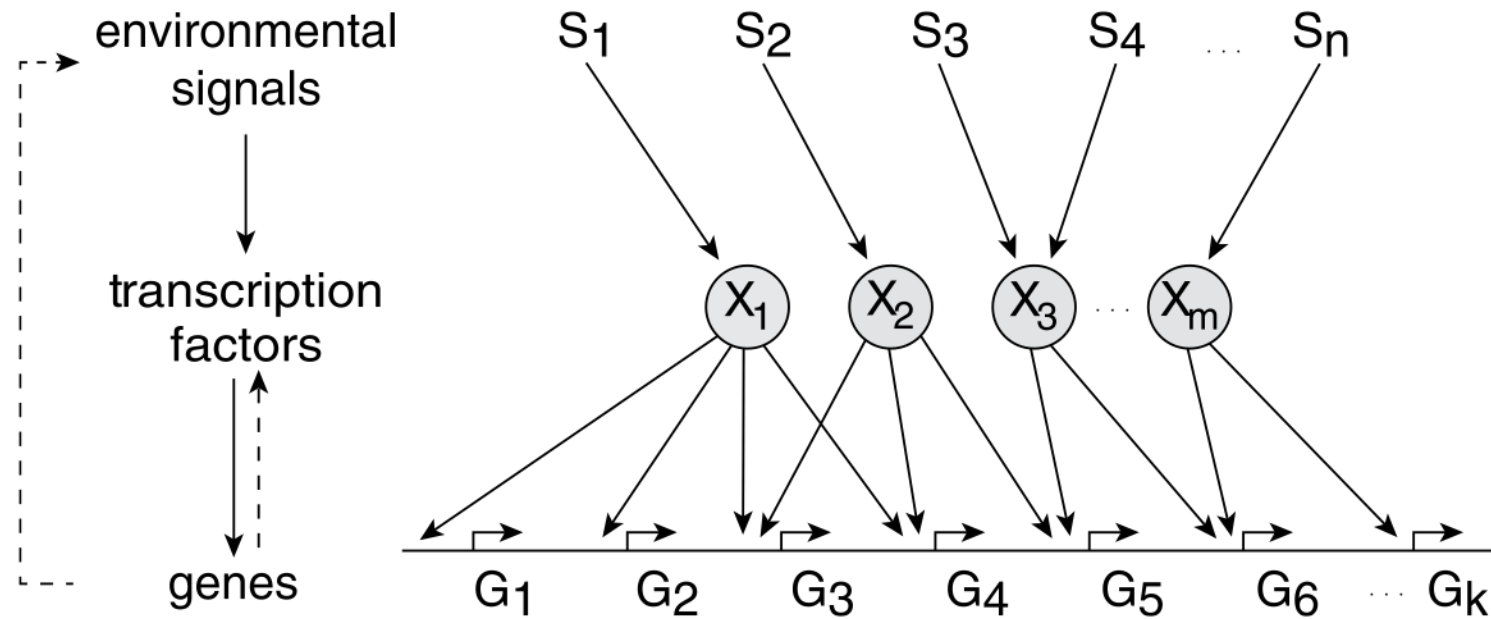
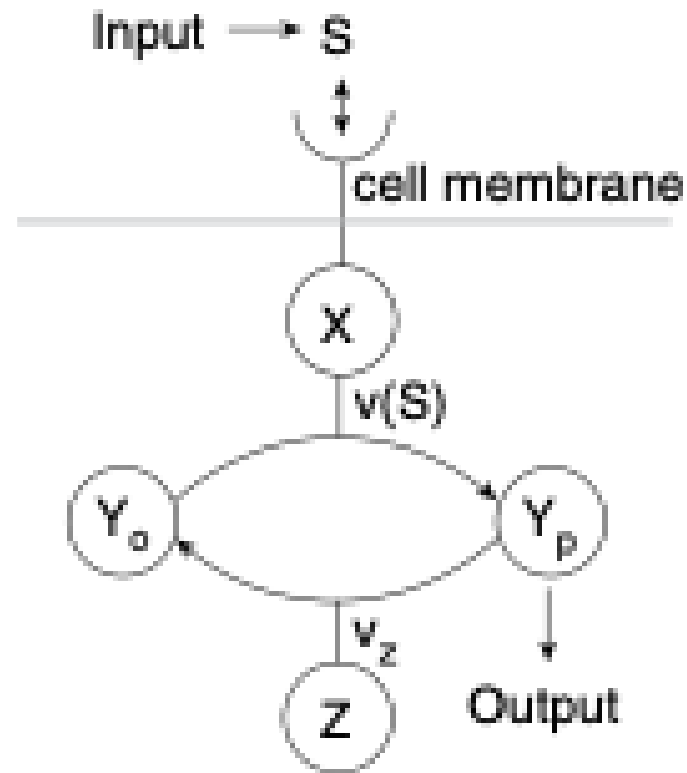


Lecture 1 Chemotaxis

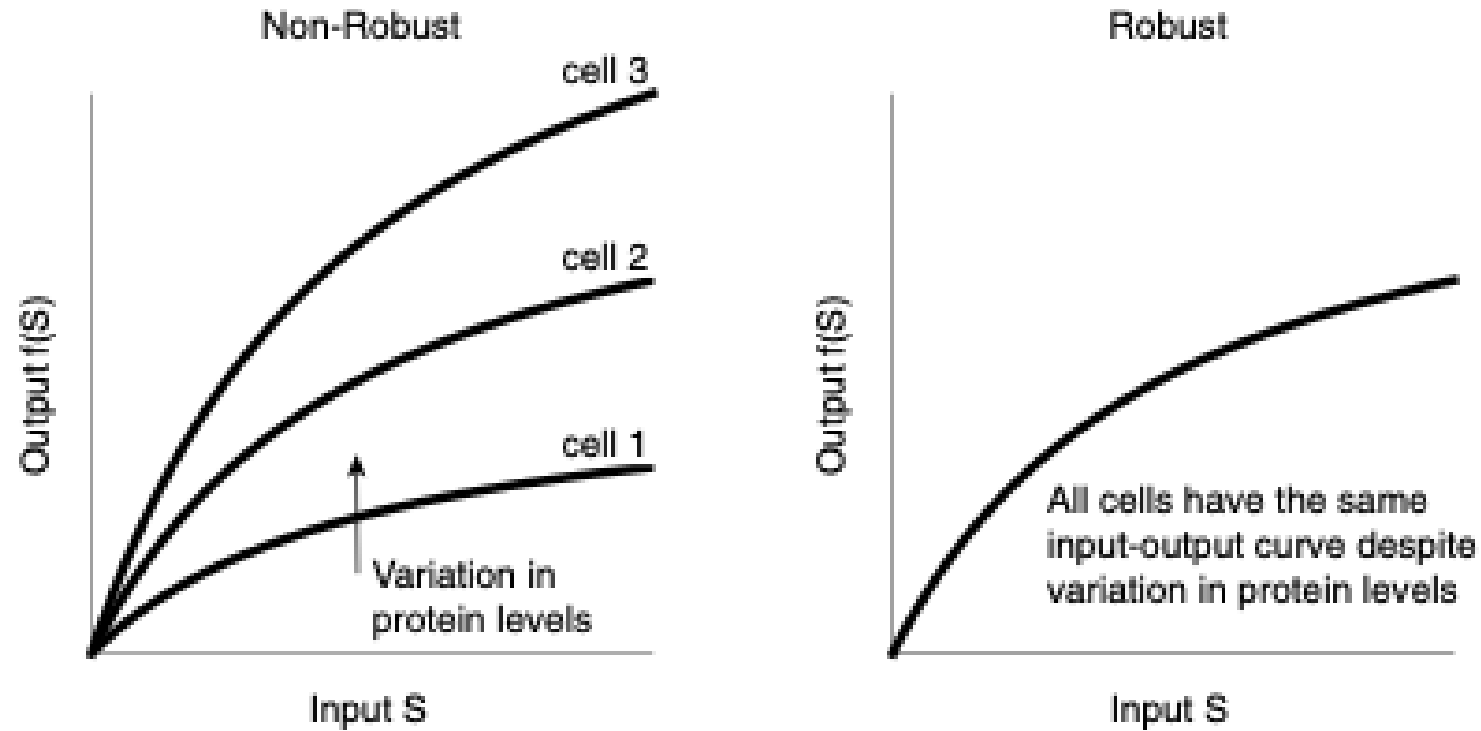
How do cells think ?



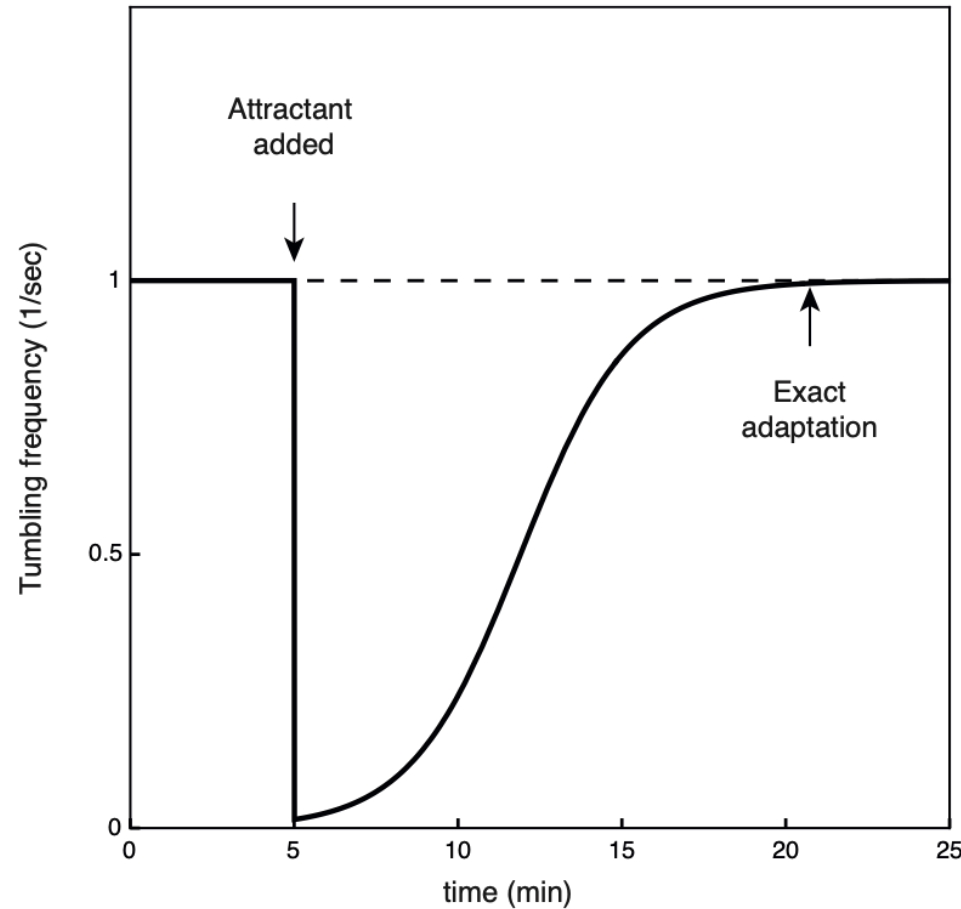
Simple models of signal-transduction circuits are not robust



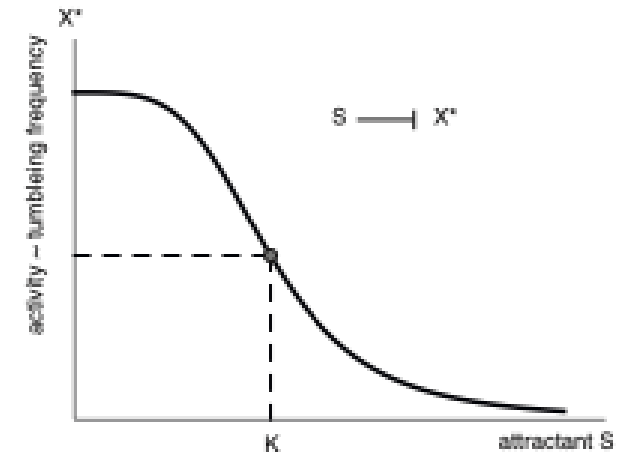
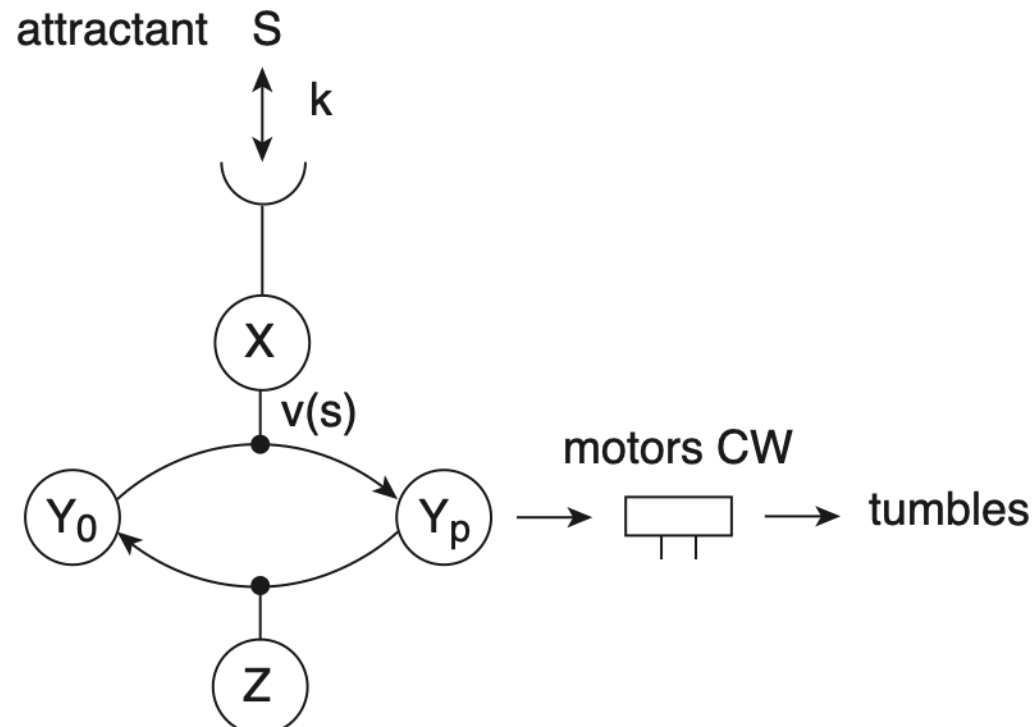
Biological circuits must operate robustly in the face of variations



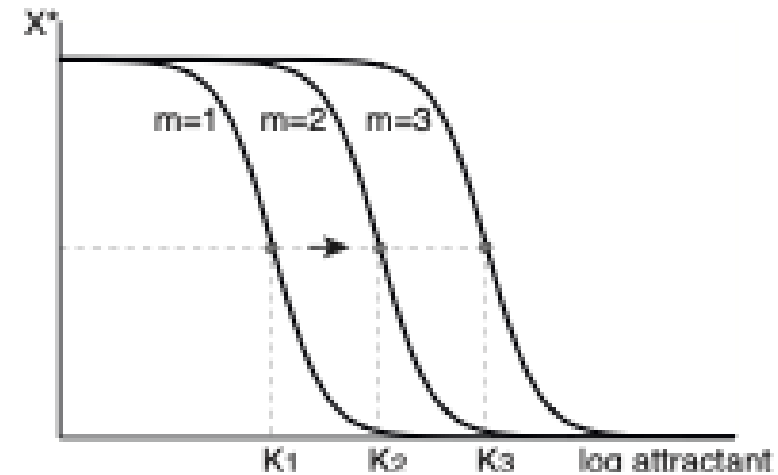
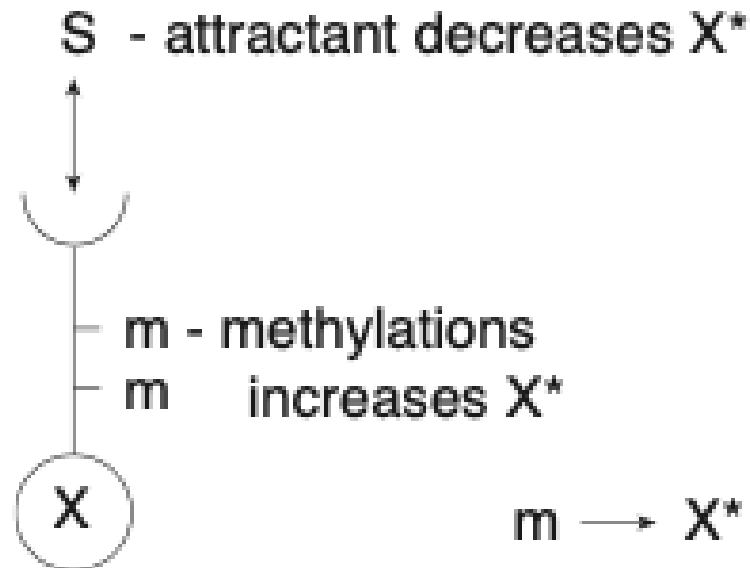
A basic feature of the chemotaxis response can be revealed by a simple experiment



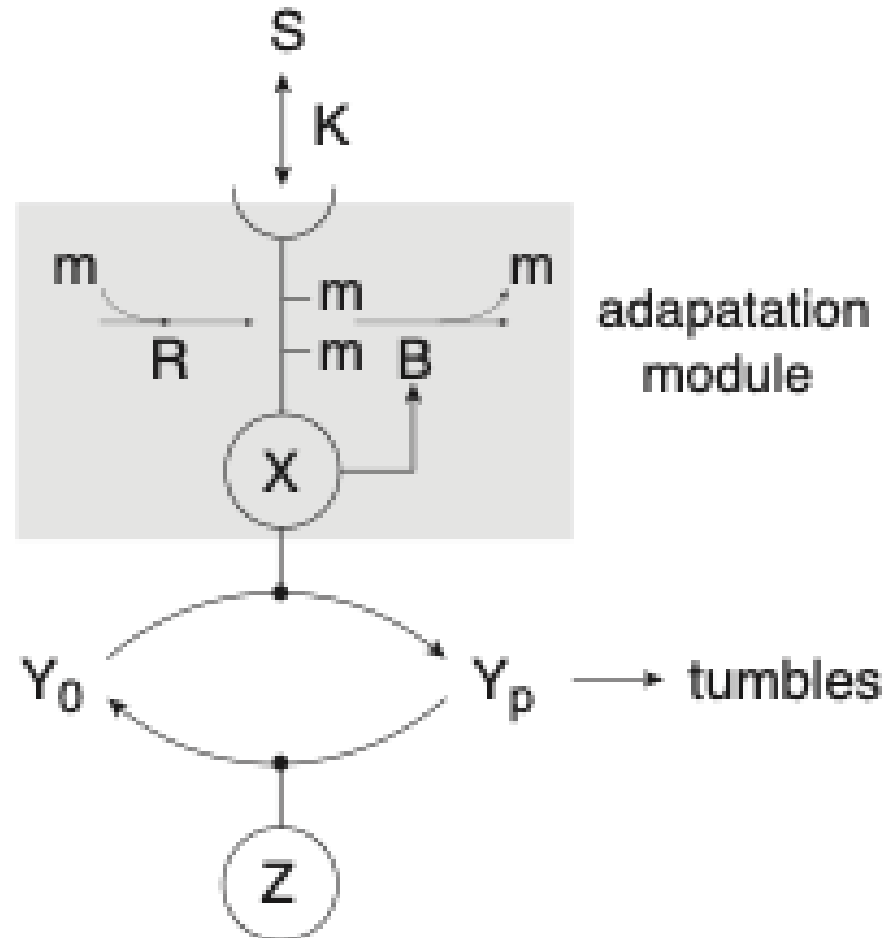
Attractant lowers the activity of X



Adaptation is due to slow modification of X that increases its activity

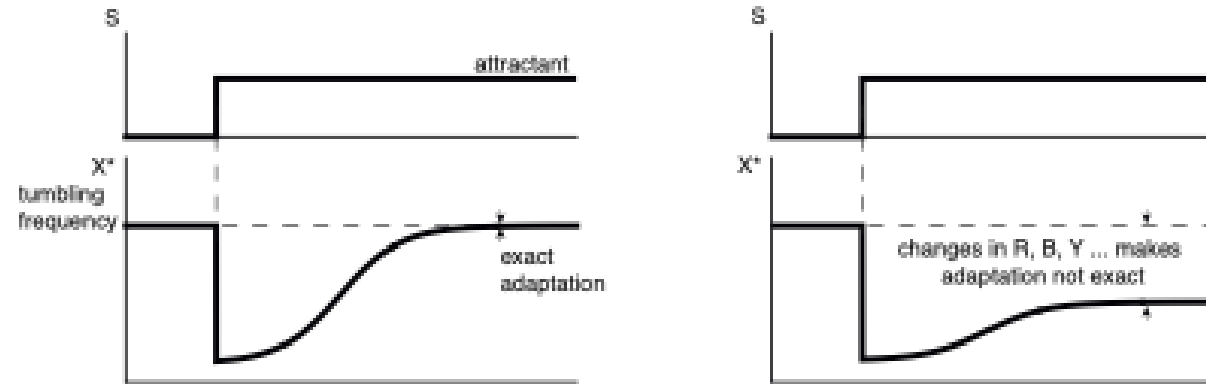


CheR and CheB form an adaptation module

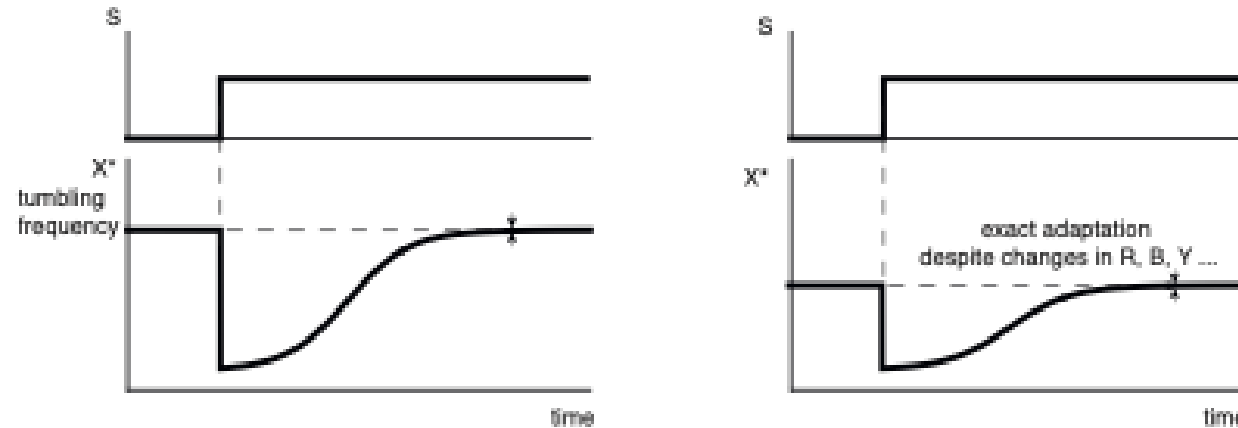


Many model adapt – but only some do so robustly

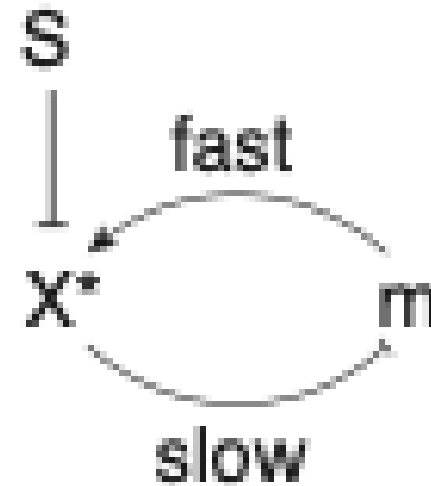
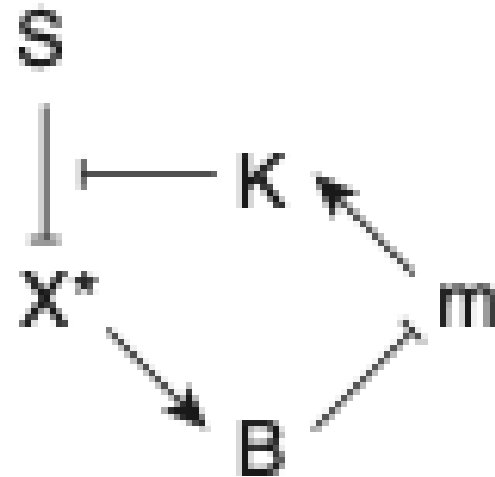
Non robust model



Robust model



Barkai–Leibler's key proposal: B demethylates only active X, forming a slow negative feedback loop

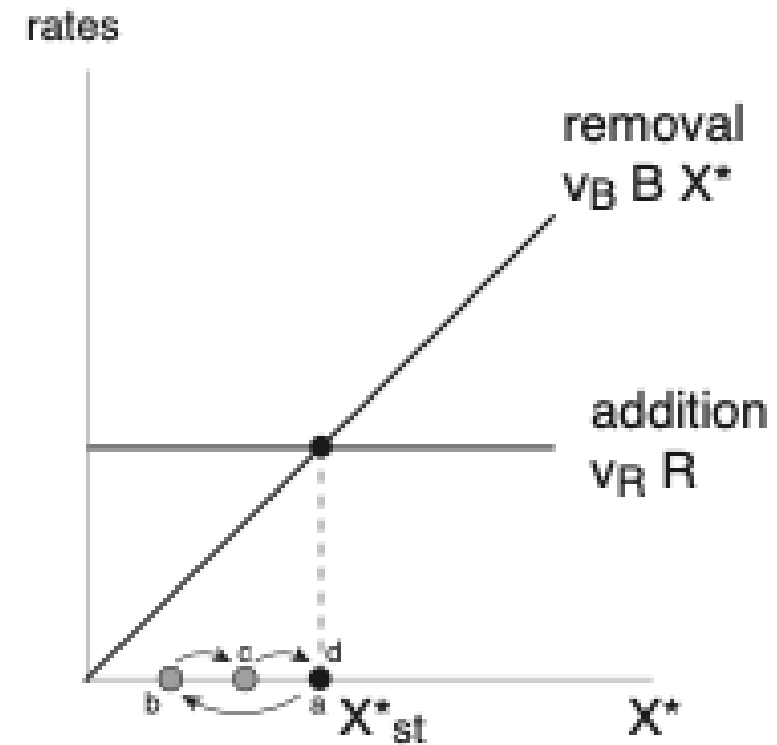
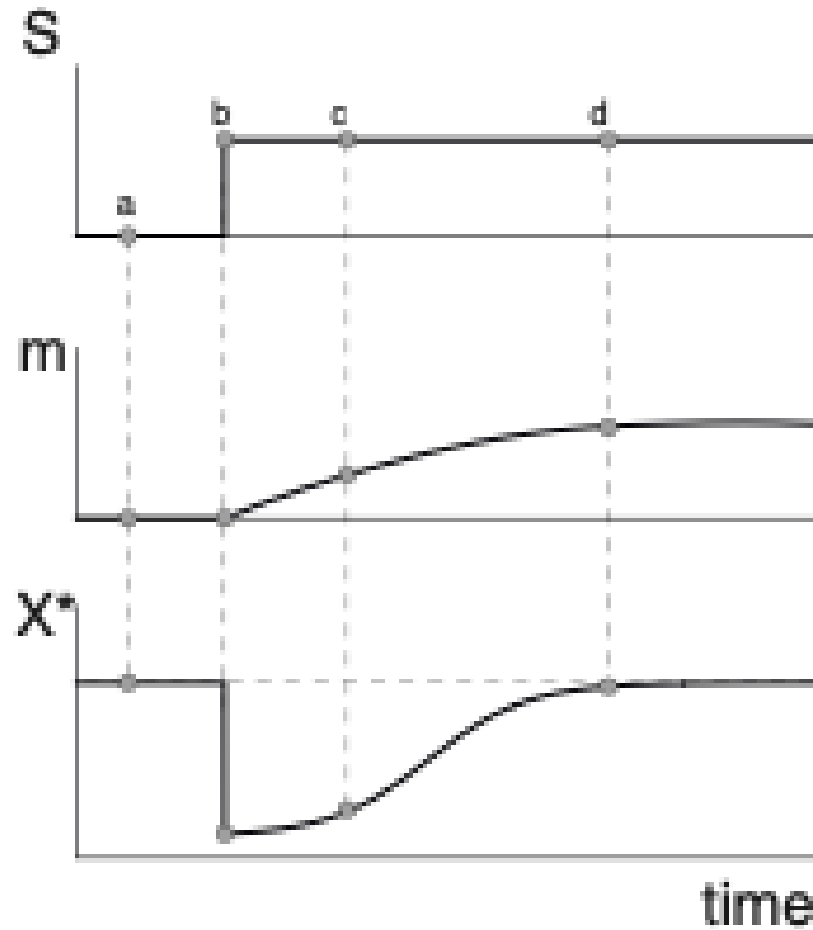


The Barkai-Leibler model explains robust exact adaptation

$$dm/dt = V_r R - V_b B X^*$$

$$X^*_{st} = V_r R / V_b B.$$

Exact adaptation is robust; baseline activity and adaptation time are not



Barkai-Leibler model is mapped to integral feedback

$$\frac{dm}{dt} = VbB(X_{st}^* - X^*)$$

$$m(t) \sim \int_{-\infty}^t (X_{st}^* - X^*(t)) dt \sim \int error(t) dt$$

Alon et al. experimentally test robust exact adaptation

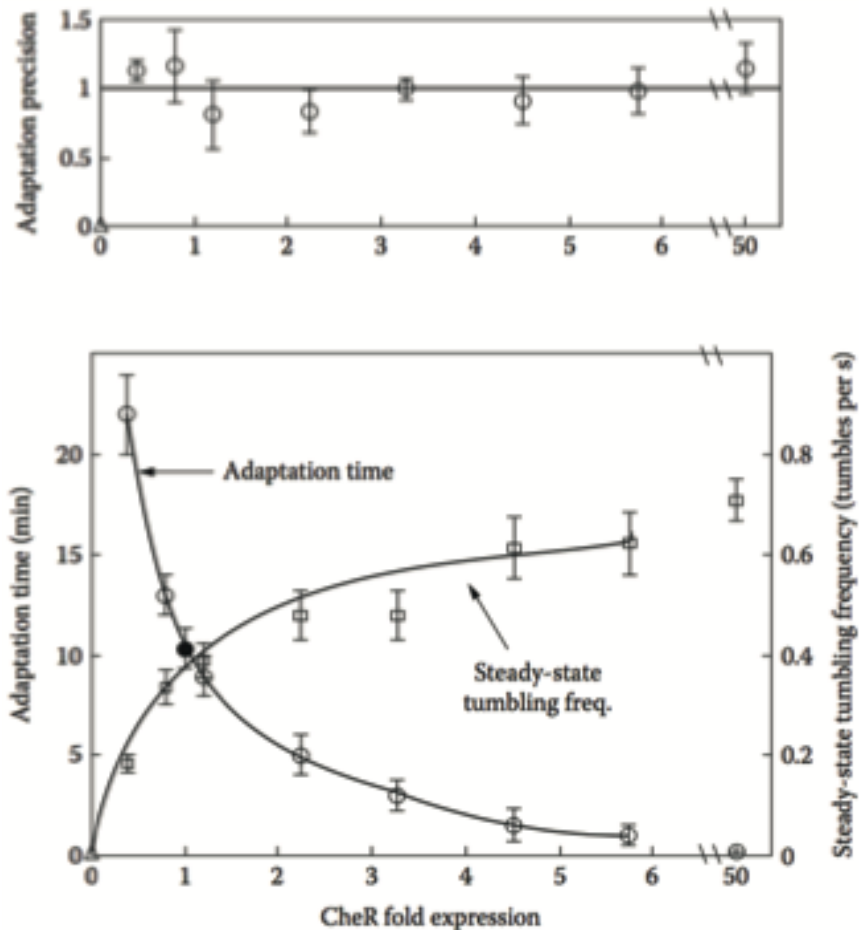
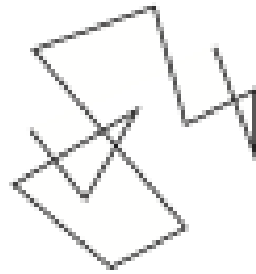


Table 1 Effect of variation in chemotaxis protein levels on behaviour

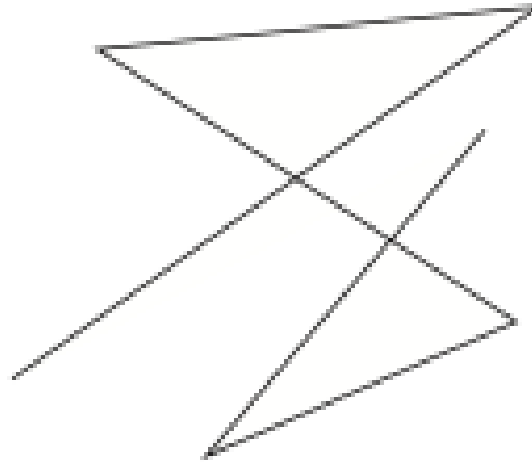
Protein varied	Fold expression	Strain background	Steady-state tumbling frequency (s^{-1})	Adaptation time (min)	Precision of adaptation
Wild type	1.0	Wild type (RP437)	0.44 ± 0.03	10 ± 1	0.98 ± 0.05
CheB*	0.4 ± 0.1	$\Delta cheB$ (RP4872)	0.66 ± 0.06	7 ± 1	0.98 ± 0.12
CheB*	12 ± 3	$\Delta cheB$ (RP4872)	0.34 ± 0.02	15 ± 1	1.09 ± 0.11
CheB Δ 1	~ 11	$\Delta cheB$ (RP4872)	0.74 ± 0.06	9 ± 2	0.90 ± 0.13
CheY*	0.2 ± 0.1	$\Delta cheY$ (RP6231)	0.24 ± 0.04	11 ± 3	1.04 ± 0.08
CheZ	0	$\Delta cheZ$ (RP1616)	1.6 ± 0.1	10 ± 2	1.1 ± 0.34
Tar, Tap, CheR, B, Y, Z	5 ± 2	Wild-type (RP437)	0.30 ± 0.06	3 ± 1	1.04 ± 0.07

Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)

Protein concentration fluctuations result in individuality in chemotaxis



rapid progress
in dense obstacles



rapid progress
in free liquid